POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6043D | Semester VI | Theory | 4 Credits

Principles of MEMS

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6043D as Principles of MEMS in Semester VI for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned "Requested file not found." With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur's MEMS and Microsystems course and polytechnic MEMS curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Microsystem designs, fabrication recipes, sensor calibrations and material characterizations must be based on approved engineering plans, certified cleanroom data, current safety standards and competent professional review.

Verified source note: Verified sources support scaling laws, microfabrication (LIGA, etching), MEMS sensors, and actuators. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Principles of MEMS (Micro-Electro-Mechanical Systems) studies the integration of mechanical elements, sensors, actuators and electronics on a common silicon substrate. It develops the ability to analyze scaling effects at the micro-scale, evaluate microfabrication techniques like bulk and surface micromachining, understand the operation of MEMS sensors and actuators, and coordinate microsystem solutions while respecting the technical and material boundaries of electronic and mechanical engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic electronics, physics, chemistry, materials science and basic mechanical principles.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the fundamentals of MEMS, microsystems and scaling laws in miniaturization.
2. Analyze the materials and microfabrication processes used for MEMS devices.
3. Evaluate the principles and operation of MEMS sensors (pressure, acceleration, etc.).
4. Explain the systematic design of MEMS actuators and the basics of microfluidics.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the fundamentals of MEMS and scaling effects in miniaturization.	Understand
CO2	Analyze microfabrication techniques and materials for MEMS devices.	Apply
CO3	Analyze the characteristics and applications of MEMS sensors.	Apply
CO4	Explain the operation of MEMS actuators and microfluidic systems.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Introduction to MEMS and Scaling Laws	Definition of MEMS and Microsystems; History and Evolution; Applications in consumer and industrial electronics; Scaling laws: Scaling in geometry, Electrostatic forces, Electromagnetic forces, and Heat transfer.	Approx. 14 hours	CO1	Module 1
2	MEMS Materials and Microfabrication	Materials for MEMS: Silicon, Polymers, Metals; Microfabrication: Photolithography, Ion implantation, Diffusion; Etching: Wet and Dry etching; Bulk and Surface micromachining; LIGA process.	Approx. 16 hours	CO2	Module 2
3	MEMS Sensors and Applications	Sensing principles: Piezoresistive, Capacitive, Piezoelectric; MEMS Pressure sensors; MEMS Accelerometers: Piezoresistive and Capacitive types; MEMS Gyroscopes.	Approx. 16 hours	CO3	Module 3
4	MEMS Actuators and Microfluidics	Actuation principles: Electrostatic, Thermal, Piezoelectric; MEMS Micro-mirrors; Inkjet printer heads; Microfluidics: Flow at the micro-scale, Micro-pumps, Lab-on-a-Chip basics.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Introduction to MEMS and Scaling Laws

Definition of MEMS and Microsystems; History and Evolution; Applications in consumer and industrial electronics; Scaling laws: Scaling in geometry, Electrostatic forces, Electromagnetic forces, and Heat transfer.

CO1 · Approx. 14 hours

MEMS Materials and Microfabrication

Materials for MEMS: Silicon, Polymers, Metals; Microfabrication: Photolithography, Ion implantation, Diffusion; Etching: Wet and Dry etching; Bulk and Surface micromachining; LIGA process.

CO2 · Approx. 16 hours

MEMS Sensors and Applications

Sensing principles: Piezoresistive, Capacitive, Piezoelectric; MEMS Pressure sensors; MEMS Accelerometers: Piezoresistive and Capacitive types; MEMS Gyroscopes.

CO3 · Approx. 16 hours

MEMS Actuators and Microfluidics

Actuation principles: Electrostatic, Thermal, Piezoelectric; MEMS Micro-mirrors; Inkjet printer heads; Microfluidics: Flow at the micro-scale, Micro-pumps, Lab-on-a-Chip basics.

CO4 · Approx. 14 hours

MODULE 1

Introduction to MEMS and Scaling Laws

Definition of MEMS and Microsystems; History and Evolution; Applications in consumer and industrial electronics; Scaling laws: Scaling in geometry, Electrostatic forces, Electromagnetic forces, and Heat transfer.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

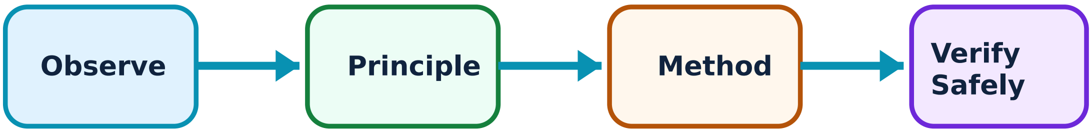
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Introduction to MEMS and Scaling Laws: identify the system, state its principle, follow the correct method and verify the result safely.

1. MEMS Fundamentals−

MEMS integrates mechanical and electrical components at the micro-scale. Microsystems include the sensing/actuating element and the interface electronics. Advantages include miniaturization, low power, and mass production. Common applications include smartphone sensors and automotive safety systems.

Study and application method

Define 'MEMS' and 'Microsystem'; list three applications of MEMS in modern electronics.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'MEMS' and 'Microsystem'; list three applications of MEMS in modern electronics.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: MEMS devices are highly sensitive to environmental conditions. Do not handle exposed MEMS chips without authorised ESD protection and dust-free environments.

2. Scaling in Miniaturization−

Scaling laws describe how physical parameters change as dimensions are reduced. As size decreases, surface-area-to-volume ratio increases, making surface forces (like electrostatics and surface tension) dominant over volume forces (like gravity and inertia).

Study and application method

Explain why 'Electrostatic' forces are preferred over 'Electromagnetic' forces at the micro-scale; discuss the impact of scaling on 'Heat Transfer'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain why 'Electrostatic' forces are preferred over 'Electromagnetic' forces at the micro-scale; discuss the impact of scaling on 'Heat Transfer'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Scaling can lead to unexpected effects like 'Stiction' (static friction). All MEMS designs must follow authorised surface-treatment and anti-stiction strategies.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

MEMS Materials and Microfabrication

Materials for MEMS: Silicon, Polymers, Metals; Microfabrication: Photolithography, Ion implantation, Diffusion; Etching: Wet and Dry etching; Bulk and Surface micromachining; LIGA process.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

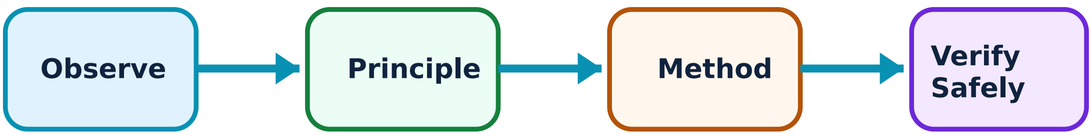
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for MEMS Materials and Microfabrication: identify the system, state its principle, follow the correct method and verify the result safely.

1. MEMS Materials−

Silicon is the primary material due to its excellent mechanical properties and existing fabrication infrastructure. Polymers (like SU-8) are used for microfluidics. Piezoelectric materials (like PZT) are used for sensing and actuation.

Study and application method

Explain why 'Silicon' is an ideal material for MEMS; compare 'Silicon' and 'Polymers' for microsystem applications.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain why 'Silicon' is an ideal material for MEMS; compare 'Silicon' and 'Polymers' for microsystem applications.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Material properties at the micro-scale differ from bulk values. Do not use macro-scale material constants for MEMS design without authorised experimental verification.

2. Micromachining Techniques−

Bulk micromachining carves structures into the silicon substrate using anisotropic etching. Surface micromachining builds structures on top of the substrate using sacrificial layers. LIGA (Lithography, Electroplating, and Molding) is used for high-aspect-ratio structures.

Study and application method

Differentiate between 'Bulk' and 'Surface' micromachining with conceptual sketches; describe the 'LIGA' process steps.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Bulk' and 'Surface' micromachining with conceptual sketches; describe the 'LIGA' process steps.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Etching chemicals (like HF or KOH) are highly hazardous. All fabrication must follow authorised chemical handling and waste disposal protocols.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

MEMS Sensors and Applications

Sensing principles: Piezoresistive, Capacitive, Piezoelectric; MEMS Pressure sensors; MEMS Accelerometers: Piezoresistive and Capacitive types; MEMS Gyroscopes.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

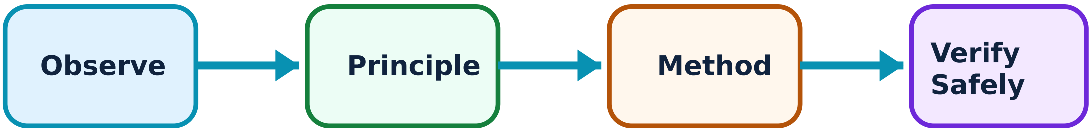
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for MEMS Sensors and Applications: identify the system, state its principle, follow the correct method and verify the result safely.

1. Sensing Principles−

MEMS sensors convert physical stimuli into electrical signals. Piezoresistive sensing uses changes in resistance under strain. Capacitive sensing uses changes in capacitance due to displacement. Piezoelectric sensing generates charge under mechanical stress.

Study and application method

Compare 'Piezoresistive' and 'Capacitive' sensing in a conceptual table; explain the 'Piezoelectric Effect' in MEMS.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Piezoresistive' and 'Capacitive' sensing in a conceptual table; explain the 'Piezoelectric Effect' in MEMS.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Sensor cross-sensitivity (e.g., pressure sensor responding to temperature) can cause errors. All sensor designs must follow authorised compensation and calibration procedures.

2. Inertial and Pressure Sensors−

MEMS pressure sensors use diaphragms that deflect under pressure. Accelerometers use a proof mass and springs to detect motion. Gyroscopes use the Coriolis effect to measure angular velocity. These are critical for stabilization and navigation.

Study and application method

Sketch the structure of a 'MEMS Accelerometer' conceptually; explain how a 'MEMS Gyroscope' utilizes the Coriolis effect.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the structure of a 'MEMS Accelerometer' conceptually; explain how a 'MEMS Gyroscope' utilizes the Coriolis effect.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Inertial sensors are prone to bias drift and noise. Do not use raw MEMS sensor data for critical navigation without authorised filtering (e.g., Kalman filter) and sensor-fusion algorithms.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

MEMS Actuators and Microfluidics

Actuation principles: Electrostatic, Thermal, Piezoelectric; MEMS Micro-mirrors; Inkjet printer heads; Microfluidics: Flow at the micro-scale, Micro-pumps, Lab-on-a-Chip basics.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

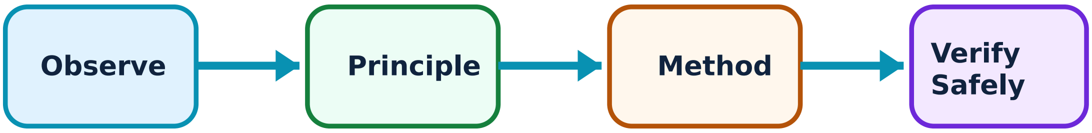
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for MEMS Actuators and Microfluidics: identify the system, state its principle, follow the correct method and verify the result safely.

1. MEMS Actuators−

Actuators convert electrical signals into mechanical movement. Electrostatic actuators use attractive forces between plates. Thermal actuators use differential expansion. Micro-mirrors (DMD) are used in digital projectors for high-speed light switching.

Study and application method

Describe the working of an 'Electrostatic Comb-drive' actuator; explain the application of MEMS in 'Digital Micromirror Devices' (DMD).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working of an 'Electrostatic Comb-drive' actuator; explain the application of MEMS in 'Digital Micromirror Devices' (DMD).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Common mistake / precaution: High voltages in electrostatic actuators can cause 'Pull-in' failure. All actuator designs must follow authorised voltage-limit and gap-spacing analysis.

2. Microfluidics and Lab-on-a-Chip−

Microfluidics deals with the behavior and control of fluids at the micro-scale. Flow is typically laminar (low Reynolds number). Lab-on-a-Chip (LOC) integrates multiple laboratory functions on a single chip for rapid medical diagnostics and chemical analysis.

Study and application method

Define 'Laminar Flow' in microfluidics; list the components of a 'Lab-on-a-Chip' system conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Laminar Flow' in microfluidics; list the components of a 'Lab-on-a-Chip' system conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Microfluidic channels can easily clog due to particles or bubbles. All fluidic designs must follow authorised filtration and surface-wetting protocols to ensure reliable operation.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Introduction to MEMS and Scaling Laws

Use the concept flow to revise observation, principle, method and verification.

Module 2: MEMS Materials and Microfabrication

Use the concept flow to revise observation, principle, method and verification.

Module 3: MEMS Sensors and Applications

Use the concept flow to revise observation, principle, method and verification.

Module 4: MEMS Actuators and Microfluidics

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Bulk and Surface micromachining in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the 'Coriolis Force' acting on a vibrating proof mass in a MEMS gyroscope.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the structure of a MEMS Capacitive Pressure Sensor showing the diaphragm and electrodes.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the 'Scaling Factor' conceptually for the electrostatic force when dimensions are reduced by 10.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a MEMS device in your smartphone (e.g., accelerometer, microphone) and note its benefits.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Introduction to MEMS and Scaling Laws

Explain MEMS Fundamentals and state one correct method, application or precaution.−

MEMS integrates mechanical and electrical components at the micro-scale. Microsystems include the sensing/actuating element and the interface electronics. Advantages include miniaturization, low power, and mass production. Common applications include smartphone sensors and automotive safety systems.

Answer framework: Define 'MEMS' and 'Microsystem'; list three applications of MEMS in modern electronics.

Important point: MEMS devices are highly sensitive to environmental conditions. Do not handle exposed MEMS chips without authorised ESD protection and dust-free environments.

Explain Scaling in Miniaturization and state one correct method, application or precaution.−

Scaling laws describe how physical parameters change as dimensions are reduced. As size decreases, surface-area-to-volume ratio increases, making surface forces (like electrostatics and surface tension) dominant over volume forces (like gravity and inertia).

Answer framework: Explain why 'Electrostatic' forces are preferred over 'Electromagnetic' forces at the micro-scale; discuss the impact of scaling on 'Heat Transfer'.

Important point: Scaling can lead to unexpected effects like 'Stiction' (static friction). All MEMS designs must follow authorised surface-treatment and anti-stiction strategies.

Module 2 — MEMS Materials and Microfabrication

Explain MEMS Materials and state one correct method, application or precaution.−

Silicon is the primary material due to its excellent mechanical properties and existing fabrication infrastructure. Polymers (like SU-8) are used for microfluidics. Piezoelectric materials (like PZT) are used for sensing and actuation.

Answer framework: Explain why 'Silicon' is an ideal material for MEMS; compare 'Silicon' and 'Polymers' for microsystem applications.

Important point: Material properties at the micro-scale differ from bulk values. Do not use macro-scale material constants for MEMS design without authorised experimental verification.

Explain Micromachining Techniques and state one correct method, application or precaution.−

Bulk micromachining carves structures into the silicon substrate using anisotropic etching. Surface micromachining builds structures on top of the substrate using sacrificial layers. LIGA (Lithography, Electroplating, and Molding) is used for high-aspect-ratio structures.

Answer framework: Differentiate between 'Bulk' and 'Surface' micromachining with conceptual sketches; describe the 'LIGA' process steps.

Important point: Etching chemicals (like HF or KOH) are highly hazardous. All fabrication must follow authorised chemical handling and waste disposal protocols.

Module 3 — MEMS Sensors and Applications

Explain Sensing Principles and state one correct method, application or precaution.−

MEMS sensors convert physical stimuli into electrical signals. Piezoresistive sensing uses changes in resistance under strain. Capacitive sensing uses changes in capacitance due to displacement. Piezoelectric sensing generates charge under mechanical stress.

Answer framework: Compare 'Piezoresistive' and 'Capacitive' sensing in a conceptual table; explain the 'Piezoelectric Effect' in MEMS.

Important point: Sensor cross-sensitivity (e.g., pressure sensor responding to temperature) can cause errors. All sensor designs must follow authorised compensation and calibration procedures.

Explain Inertial and Pressure Sensors and state one correct method, application or precaution.−

MEMS pressure sensors use diaphragms that deflect under pressure. Accelerometers use a proof mass and springs to detect motion. Gyroscopes use the Coriolis effect to measure angular velocity. These are critical for stabilization and navigation.

Answer framework: Sketch the structure of a 'MEMS Accelerometer' conceptually; explain how a 'MEMS Gyroscope' utilizes the Coriolis effect.

Important point: Inertial sensors are prone to bias drift and noise. Do not use raw MEMS sensor data for critical navigation without authorised filtering (e.g., Kalman filter) and sensor-fusion algorithms.

Module 4 — MEMS Actuators and Microfluidics

Explain MEMS Actuators and state one correct method, application or precaution.−

Actuators convert electrical signals into mechanical movement. Electrostatic actuators use attractive forces between plates. Thermal actuators use differential expansion. Micro-mirrors (DMD) are used in digital projectors for high-speed light switching.

Answer framework: Describe the working of an 'Electrostatic Comb-drive' actuator; explain the application of MEMS in 'Digital Micromirror Devices' (DMD).

Important point: High voltages in electrostatic actuators can cause 'Pull-in' failure. All actuator designs must follow authorised voltage-limit and gap-spacing analysis.

Explain Microfluidics and Lab-on-a-Chip and state one correct method, application or precaution.−

Microfluidics deals with the behavior and control of fluids at the micro-scale. Flow is typically laminar (low Reynolds number). Lab-on-a-Chip (LOC) integrates multiple laboratory functions on a single chip for rapid medical diagnostics and chemical analysis.

Answer framework: Define 'Laminar Flow' in microfluidics; list the components of a 'Lab-on-a-Chip' system conceptually.

Important point: Microfluidic channels can easily clog due to particles or bubbles. All fluidic designs must follow authorised filtration and surface-wetting protocols to ensure reliable operation.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Introduction to MEMS and Scaling Laws.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: MEMS Materials and Microfabrication.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: MEMS Sensors and Applications.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: MEMS Actuators and Microfluidics.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition of MEMS and Microsystems; History and Evolution; Applications in consumer and industrial electronics; Scaling laws: Scaling in geometry, Electrostatic forces, Electromagnetic forces, and Heat transfer..
2. Develop a complete answer or practical plan for: Materials for MEMS: Silicon, Polymers, Metals; Microfabrication: Photolithography, Ion implantation, Diffusion; Etching: Wet and Dry etching; Bulk and Surface micromachining; LIGA process..
3. Develop a complete answer or practical plan for: Sensing principles: Piezoresistive, Capacitive, Piezoelectric; MEMS Pressure sensors; MEMS Accelerometers: Piezoresistive and Capacitive types; MEMS Gyroscopes..
4. Develop a complete answer or practical plan for: Actuation principles: Electrostatic, Thermal, Piezoelectric; MEMS Micro-mirrors; Inkjet printer heads; Microfluidics: Flow at the micro-scale, Micro-pumps, Lab-on-a-Chip basics..

LAST-MILE REVISION

Quick Revision

Module 1: Introduction to MEMS and Scaling Laws

Definition of MEMS and Microsystems; History and Evolution; Applications in consumer and industrial electronics; Scaling laws: Scaling in geometry, Electrostatic forces, Electromagnetic forces, and Heat transfer.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: MEMS Materials and Microfabrication

Materials for MEMS: Silicon, Polymers, Metals; Microfabrication: Photolithography, Ion implantation, Diffusion; Etching: Wet and Dry etching; Bulk and Surface micromachining; LIGA process.

- Match each topic to CO2.

- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: MEMS Sensors and Applications

Sensing principles: Piezoresistive, Capacitive, Piezoelectric; MEMS Pressure sensors; MEMS Accelerometers: Piezoresistive and Capacitive types; MEMS Gyroscopes.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: MEMS Actuators and Microfluidics

Actuation principles: Electrostatic, Thermal, Piezoelectric; MEMS Micro-mirrors; Inkjet printer heads; Microfluidics: Flow at the micro-scale, Micro-pumps, Lab-on-a-Chip basics.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
MEMS Fundamentals	MEMS integrates mechanical and electrical components at the micro-scale.
Scaling in Miniaturization	Scaling laws describe how physical parameters change as dimensions are reduced.
MEMS Materials	Silicon is the primary material due to its excellent mechanical properties and existing fabrication infrastructure.
Micromachining Techniques	Bulk micromachining carves structures into the silicon substrate using anisotropic etching.
Sensing Principles	MEMS sensors convert physical stimuli into electrical signals.
Inertial and Pressure Sensors	MEMS pressure sensors use diaphragms that deflect under pressure.
MEMS Actuators	Actuators convert electrical signals into mechanical movement.
Microfluidics and Lab-on-a-Chip	Microfluidics deals with the behavior and control of fluids at the micro-scale.

LEARNING RESOURCES

References

Recommended MEMS references

1. Tai-Ran Hsu, MEMS and Microsystems: Design and Manufacture.
2. Stephen D. Senturia, Microsystem Design.
3. Marc J. Madou, Fundamentals of Microfabrication: The Science of Miniaturization.
4. G. K. Ananthasuresh et al., Micro and Smart Systems.

Approved public MEMS sources

- http://digimat.in/nptel/courses/video/117105082/L01.html
- https://nptel.ac.in/courses/108106165

These sources provide the verified study basis while official Course 6043D detail is unavailable; they do not replace official chip designs, authorised cleanroom protocols, certified equipment specifications or authorised research plans.